

What Cycle Closure Can and Cannot See in a Relative Binding Free Energy Network

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Abstract

Cycle closure is the standard consistency check on a perturbation network, and what such a check can see is fixed by the gradient-cycle decomposition of edge flows (HodgeRank): the cycle part of a network’s error is auditable and biases nothing, while the gradient part biases the affinities and is invisible to any statistic computed from the network’s own values, at any magnitude. Both parts are measured here on 1145 real binding edges. The auditable subspace spans a third of the error space and, in the metric the closure statistic uses, carries under one per cent of the error against experiment. Experimental and annotation error is per-ligand, so it falls in the invisible part by construction and can only have made that share smaller: the measurement is a floor, and in that metric it survives correction for that error on every system measured, at one cell as a tie.

Fitting closure against the precisions the estimator already reports turns the check into a test with a stated null, and that test behaves as the bound predicts. It ties the standard fixed 1.0 kcal/mol cutoff, and on every replicate the systems it flags lie inside the set that cutoff flags. Its per-edge residuals and its ranking of systems reproduce across independent repeats; the list of flagged systems does not. It flags nothing once the reported errors are widened by half, which is less than published estimates of the pose and preparation variance the repeats hold fixed, and it names no removal that recovers accuracy against experiment. Resolving the gradient part needs measurements from outside the network, at worst one per unresolved dimension.

1 Introduction

A structure-based drug discovery program spends most of its budget making molecules, and which molecules get made depends on how tightly each one is expected to bind its target. Predicting that affinity before the synthesis is therefore one of the load-bearing computational problems in the field, and the methods available for it trade accuracy against cost: the fast ones rank large libraries cheaply, and the accurate ones are expensive enough that each prediction has to be worth paying for.

Relative binding free energy calculations are the most accurate computational tool available for ranking how tightly a set of candidate molecules will bind a target, and structure-based drug discovery programs increasingly build them into decisions about which molecules to synthesize. The calculations come as a network rather than as isolated numbers: each edge is an alchemical transformation between two related candidates, costing hours of molecular dynamics, and a realistic project touches dozens to hundreds of them. Because the affinities a program acts on are read off

the network as a whole and not off any single edge, the network’s own internal agreement is what a practitioner has to judge the calculations by before any molecule is made.

That judgement has a standard instrument. A perturbation network must close its cycles: the signed sum of $\Delta\Delta G$ around any loop in the network is zero in expectation, and how far a network fails to close is routinely reported as a diagnostic and acted on through fixed hysteresis cutoffs. The cycles meant here are loops over several edges of the network, the object of cycle-closure analysis; the *thermodynamic* cycle is the alchemical square behind a single edge, and the two are kept apart throughout. What has not been asked of the diagnostic is how much of a network’s error it can reach at all: whether a network that closes its cycles is thereby a network whose affinities are right, and if not, what share of what is wrong with it a closure check could ever respond to.

Connectivity and redundancy answer different questions, and only the second is asked here. A connected network determines every $\Delta\Delta G$ up to one offset per component, redundant or not, which is all estimation needs. Detection is separate: it asks which errors leave a trace in the values the network reports, and the answer depends on redundancy rather than on connectivity.

That capacity follows from the algebra of the network rather than from any particular protocol, and it is measurable on public data. A network’s per-edge error splits, uniquely and orthogonally, into a part carried by its cycles and a part that is a difference of per-ligand offsets. The first is the part any closure check sees, and it does not bias the estimated affinities at all; the second is the part that biases them, and no statistic computed from the network’s own edge values responds to it at any magnitude. That fixes the capacity of a closure check; how much of a real network’s error falls inside it is then a measurement. Both are made here, on a public benchmark of real binding edges with independent protocol repeats, and both are small: the auditable subspace spans a third of the error space, and the share of the error against experiment that falls inside it is under one per cent. Figure 1 runs the whole of that from left to right.

The test that runs against this bound is built from an uncertainty the calculation already supplies rather than from one learned alongside the prediction: where the prevailing approach fits a mean-variance head [22] or reads uncertainty off the disagreement of an ensemble [17], the estimator that turns raw simulation output into a free energy for a single edge [4] has a closed-form finite-sampling variance computed from that same output at no label cost. The bar the test carries is the multistate standard error that same calculation reports, of which that closed form is the two-state case, and validating it against independent repeats is what turns cycle closure from a descriptive check into a test with a stated null.

Section 2 places the work against prior free-energy estimators, learned uncertainty and cycle-closure diagnostics. Section 3 gives the decomposition and what it fixes about any closure-based check, and Section 4 measures where a real network’s error lies against that bound. Section 5 characterises the per-edge error bar the test uses as its null and validates it against independent repeats, and Section 6 builds the test and reports what it inherits from the bound. Section 7 gives the data, the estimators, the models and the simulations behind every number reported above, and Section 8 collects the limitations, including an applicability map for the calibrated uncertainty downstream.

2 Related work

BAR [4] and MBAR [25] are the standard alchemical estimators, brought to prospective potency prediction by FEP+ [28], assessed at scale [21, 24], and carrying as their per-edge uncertainty the sandwich variance of M -estimation [11, 29]; the prevailing alternative learns that uncertainty from training examples instead [1, 14, 17, 22]. Cycle closure is long established as a network

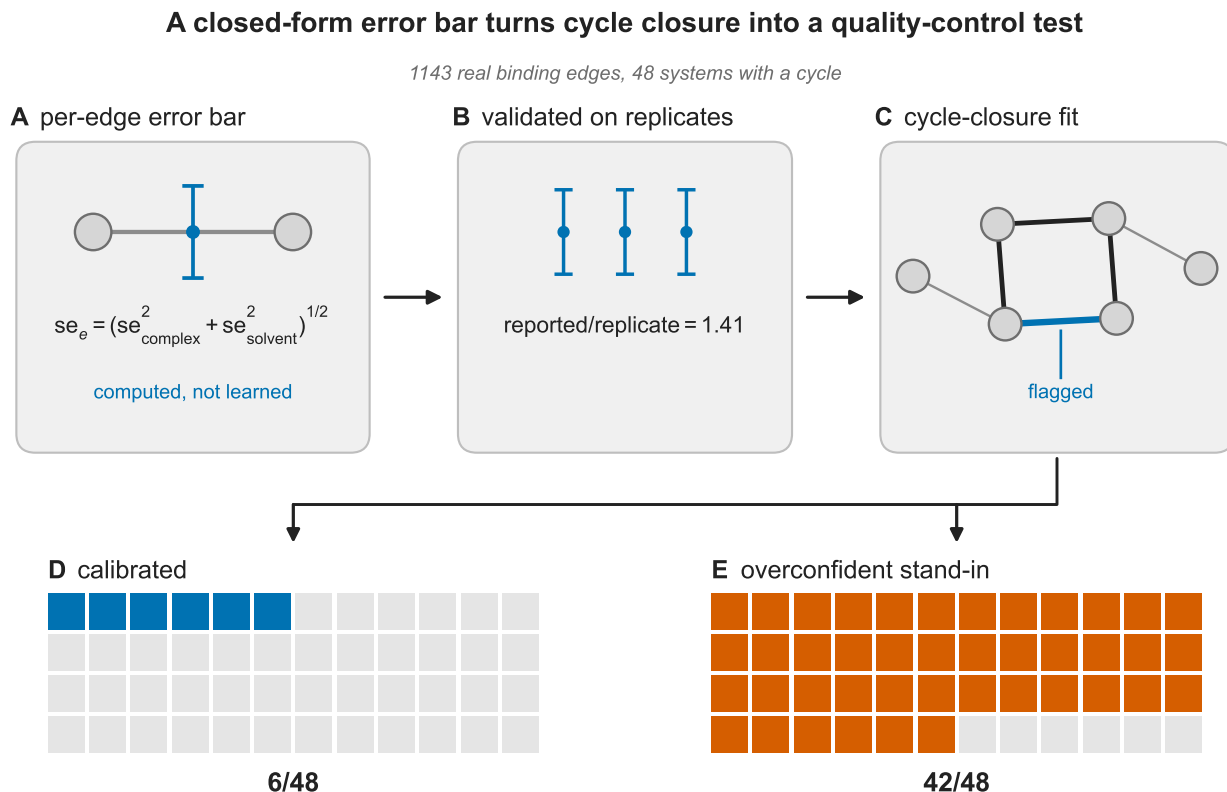


Figure 1: **Overview.** The pipeline this article runs on the benchmark, left to right: a per-edge standard error the free-energy estimator computes rather than learns, composed in quadrature from the complex-leg and solvent-leg uncertainties; its validation against the benchmark’s three independent protocol repeats; and its use as the null of a generalized-least-squares cycle-closure fit, drawn as a small perturbation network with one cycle highlighted and one edge marked as flagged.

diagnostic [5, 28], as a weighting and estimation problem [6, 18, 30], and as necessary but not sufficient [21]. The gap addressed here is the quantitative form of that insufficiency: how much of a network’s error a closure statistic can reach, and how much of a real network’s error falls there.

3 What cycle closure can see: the decomposition

A perturbation network must close its cycles: the signed sum of $\Delta\Delta G$ around any loop is zero in expectation. Everything a closure check reports is a function of the residuals of that constraint, so what such a check can see is fixed before any protocol is chosen, by the algebra of the network alone. Fix the fit those residuals come from. Node potentials are estimated by generalized least squares on the signed incidence matrix, whitened by the reported per-edge standard errors, so that the conductances are $w_e = 1/V_e^{\text{rep}}$ with V_e^{rep} the per-edge variance the free-energy estimator reports (Section 5). What that variance is on this benchmark, and when it coincides with the sandwich precision I_e^2/B_e , are Section 7 and Theorem S1 in the Supporting Information. The weighted sum of squared residuals of that fit is the closure statistic whose null Section 6 turns into a test; only its geometry is needed here. Two results follow, one per edge and one about the whole error space.

Theorem 1 (Estimation–detection conservation law). *In the whitened GLS fit let $\Pi = \mathbb{I} - H$ be the residual-maker projector onto the cycle space (Π , to keep M for M -estimation and R for a Pearson correlation). Its diagonal, the curl-leverage $h_e = \Pi_{ee}$, satisfies $h_e + w_e\Omega_e = 1$ with $w_e = 1/V_e^{\text{rep}}$ and Ω_e the effective resistance of edge e [15]; hence $\sum_e h_e = \text{dof}$, and the closure- χ^2 noncentrality against a systematic shift μ on edge e is $h_e\mu^2/V_e^{\text{rep}}$. A bridge edge (in no cycle) has $h_e = 0$ and is undetectable at any μ ; a node-consistent (gradient) bias lies in $\text{range}(H)$ and is annihilated by Π . A closure check’s blindness to node-consistent force-field bias is therefore a derived property, not an assumption, and h_e is a per-edge observability certificate with resolution $\delta_e^* = \sqrt{V_e^{\text{rep}}/h_e}$, the shift at which the closure statistic carries unit noncentrality.*

The algebra is standard: $h_e = \Pi_{ee}$ is the hat-matrix leverage of the whitened design [10], its complement $w_e\Omega_e$ is the correspondence between leverage and effective resistance that graph sparsification uses [26], and $\sum_e h_e = \text{tr}(I - H) = \text{dof}$ is the trace identity for a projector. What the theorem adds is the reading of h_e as a per-edge observability certificate, which turns the test’s blindness to node-consistent bias into a consequence rather than an assumption and names, before any simulation is run, the edges that can carry no evidence at all. The resulting map lists the structurally un-auditable (bridge) edges per system; on this benchmark the curl-leverage does not additionally predict the *location* of reproducible systematic error (studentized Spearman $\rho = -0.073$, permutation $p = 0.9571$, not significant; Supplementary Figure S9), so only the structural certificate is claimed.

Theorem 1 states the diagonal of a result about the whole error space, and that full result is what bounds the method. Write the per-edge systematic error as a vector μ over the E edges. Whitening and projecting splits it, orthogonally and uniquely, into a part that lies in the cycle space and a part that is a difference of per-ligand offsets.

Theorem 2 (Auditable and unauditable error). *In the model $y = A\varphi + \mu + \varepsilon$ with A the signed incidence matrix of a network of N ligands, E edges and c components, and φ unknown: (i) for every δ , the parameter pairs (φ, μ) and $(\varphi + \delta, \mu - A\delta)$ induce the same distribution of y , so μ is identifiable only through $\Pi\tilde{\mu}$. The auditable subspace has dimension $\text{dof} = E - N + c$ and its orthogonal complement, the gradient fields $\mu_e = b_j - b_i$, has dimension $N - c$ and is unidentifiable: no statistic computed from a network’s own edge values distinguishes a per-ligand bias from no bias, at any magnitude. (ii) The bias such an error induces in the fitted potentials, $(\tilde{A}^\top \tilde{A})^+ \tilde{A}^\top \tilde{\mu}$, depends*

only on the gradient part, while the closure noncentrality $\|\Pi\tilde{\mu}\|^2$ depends only on the cycle part. The detectable part and the biasing part are exact orthogonal complements. (iii) For an error field with no preferential alignment, the expected visible fraction $\|\Pi\tilde{\mu}\|^2/\|\tilde{\mu}\|^2$ is dof/E . (iv) Network estimation of $\Delta\Delta G$ under loop closure, with and without experimental constraints, is established [6, 30]; what the rank counting adds is the cost of the constraints. Given unbiased absolute measurements on a set S of ligands, the unidentifiable dimension is $(N - |S|) - c_{\text{free}}(S)$, with c_{free} the number of components holding none of them: the first anchor in a component removes only that component’s arbitrary offset and no unidentifiable direction, each further anchor removes exactly one dimension, and resolving a per-ligand bias in full requires an absolute measurement on every ligand.

The decomposition is not new, and none of it is claimed here. Splitting an edge field on a graph into a gradient part and a divergence-free part, and reading the size of the second as whether the first is meaningful at all, is HodgeRank [12]. The count in part (iii) is the inconsistency degrees of freedom of mixed treatment comparisons [9, 20], and the leverage machinery of part (ii) already locates inconsistency in network meta-analysis [16]. Part (i) may well be folklore in that literature, part (ii) follows from it in two lines, and part (iv) is a dimension count. What is left to this article is the reading for alchemical networks and the measurement of where a real network’s error falls between the two halves, which is Section 4 and what the rest of this article rests on. What each strand does and does not establish, and where this work is not in a position to claim priority, is set out in the Supporting Information.

The map as a design rule. The observability certificate is prospective: alone among the quantities measured here it can be acted on before a simulation is run. It prices two distinct purchases on the replicate benchmark (1143 edges over the 48 systems that carry a cycle; Section 7). Coverage costs 29 added ligand pairs, 2.5% more perturbations than were already run, and leaves no bridge anywhere; connecting every system as well costs 36. Both counts are certified minimum against the Eswaran–Tarjan augmentation bound and assume nothing about the edges they add (Supplementary Table S5; `make figDesign`).

Resolution is the second purchase, and the certificate prices it together with the point at which it stops being purchasable. An edge freshly placed on one long cycle has small h_e and correspondingly poor δ_e^* , so coverage does not buy resolution and the two are costed apart. Where a target is reachable and not already met, a greedy design reaches a median δ^* of 1.0 kcal/mol on 39 of the 48 systems for a median of two further edges, on an assumed variance for edges not yet run. Past that point the graph is no longer the binding constraint. Because $h_e \leq 1$ pointwise, no design pushes an edge’s resolution below its own reported standard error, and measuring every remaining ligand pair still leaves nine systems with a median δ^* above 1.0 kcal/mol and nineteen above 0.5. There the sampling budget is what has to change, which is the bound of Section 3 presenting itself as a design constraint rather than as a measurement.

Two properties of the certificate carry into the rule. Its targets are on a resolution scale and not a detection threshold, at the ratio Section 4 measures; and it says where the test could see, not where reproducible error lands, which the same curl-leverage does not predict.

4 Where the error actually lies

Theorem 2 makes two quantities measurable that were previously matters of assumption: how much of a network is auditable at all, and how much of its real error the audit can reach. Both are answered on this benchmark in Figure 2.

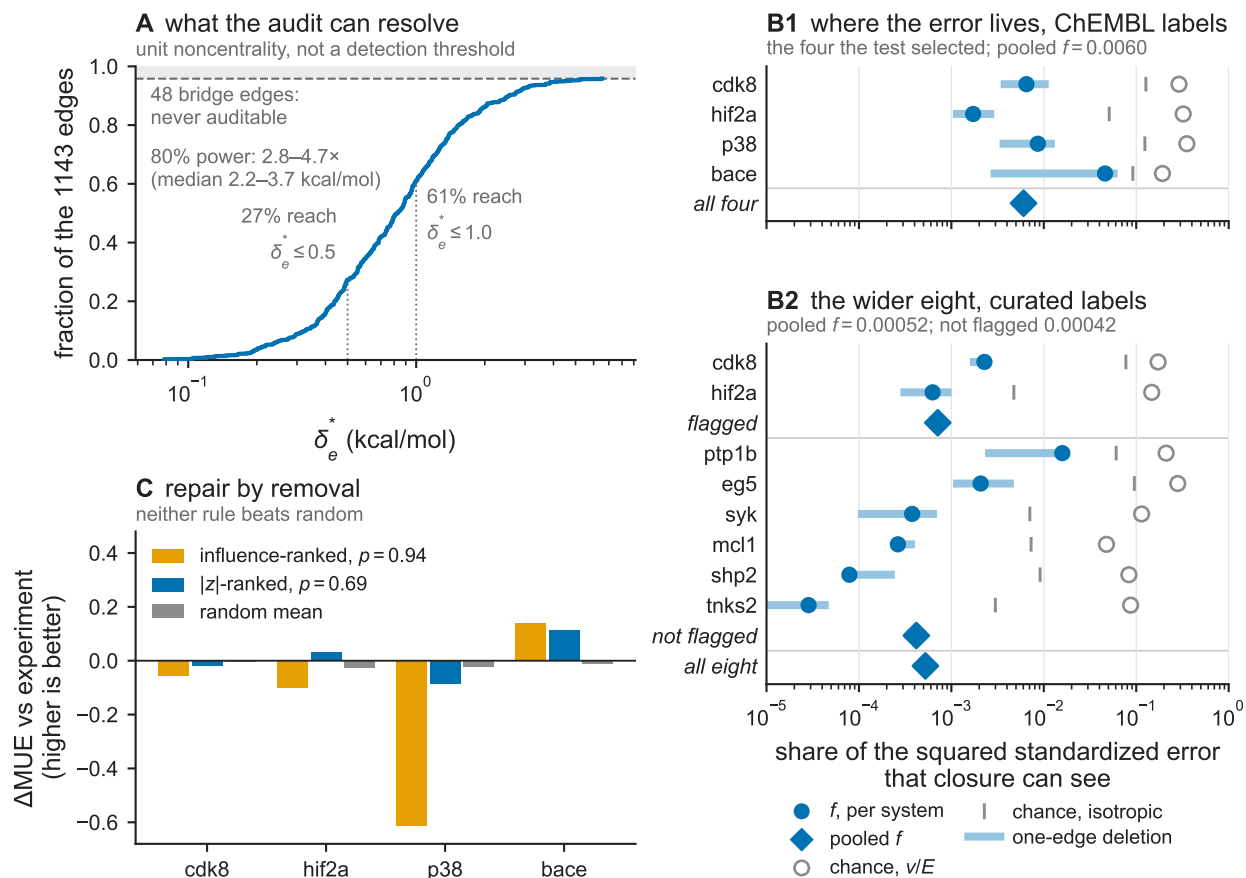


Figure 2: **Auditable capacity and the location of the error.** (A) Cumulative fraction of the 1143 benchmark edges against δ_e^* , the shift at unit noncentrality, with the bridge edges marked as the ceiling they impose. (B) The fraction f of a system’s error against experiment lying in the auditable subspace, on one shared axis, against the fraction an unaligned error field would show under two conventions, whitened-isotropic (dof/E) and isotropic in kcal/mol, and the range f takes when one edge is deleted at a time. (B1) The four sub-networks the closure test selected, grounded on a stereo-blind ChEMBL join: 167 edges, 51 residual degrees of freedom. (B2) The eight systems whose curated per-edge labels the benchmark carries and whose matched sub-network keeps a cycle: 205 edges, 31 residual degrees of freedom, grouped by whether the test flags them. Italic rows are pooled and carry f alone. (C) Change in mean unsigned error against experiment after removing the same number of edges chosen by influence, by standardized residual, and at random.

What the audit can resolve. Of the 1143 edge dimensions carried by the 48 systems with a cycle, 372 lie in the cycle space, so two-thirds of the error space is node-consistent and invisible to every closure-based diagnostic, this one included. Forty-eight edges are bridges, carrying no evidence at any magnitude. Half the remaining edges cannot resolve a shift below 0.79 kcal/mol, and only 27% of all 1143 can resolve one of 0.5 (Figure 2A). Neither figure is a detection threshold: requiring 80% power at $\alpha = 0.05$ moves that median to 2.2–3.7 kcal/mol.

That is the first of two reasons both numbers overstate the test’s reach, and the two compose. δ^* is the shift at unit noncentrality: power there runs from 0.170 to 0.070 across the degrees of freedom present, and 80% at $\alpha = 0.05$ needs 2.8 to 4.7 times it. δ^* is also carried on the raw reported standard errors, where the calibrated scale is 0.92 of them (Section 6).

Composing the two gives operational numbers worse than δ^* alone suggests. Only 6.6% of auditable edges reach 80% power against a 1.0 kcal/mol shift on that edge alone, at $\alpha = 0.05$ and before any multiplicity correction. At the level the test actually fires, a system’s most observable edge needs a median 1.35 kcal/mol for 80% power (quartiles 0.81 and 2.06) and its median edge needs 3.33; a third of systems could detect a 1.0 kcal/mol error somewhere, and 58% could detect 1.5. The regime the test operates in therefore begins at about the top of the band that relative-binding-free-energy decisions turn on, which is a sharper statement of the same limitation the rest of this section measures.

Where that capacity reaches, on four grounded sub-networks. The reach on real error is smaller than the capacity, and it is measured on different graphs. Figure 2B carries two grounded readings of that reach on one shared axis: B1 the four sub-networks the closure test itself selected, on a ChEMBL join, and B2 the eight the benchmark’s own curated per-edge labels reach, six of which the test does not flag. They run on different edges and different labels, neither corrects the other, and the differences between them are set out with the second reading below. Affinities cover part of each ligand set, so Figure 2B1 runs on each system’s grounded sub-network rather than on the full networks of Supplementary Table S4. As (N, E, c, dof) these are (24, 31, 2, 9), (41, 59, 1, 19), (34, 51, 1, 18) and (23, 26, 2, 5) for **cdk8**, **hif2a**, **p38** and **bace**, with auditable fractions 0.290, 0.322, 0.353 and 0.192 rather than the tabulated ones (**cdk8** is 0.46 on its full network); two are disconnected, which the fit absorbs into one offset per component.

All four sub-networks were selected by the calibrated closure test of Section 6, so what follows characterises the networks that test flags and not the benchmark. One of the four, **cdk8**, carries its flag from a naming collision between two releases rather than from either release’s simulations.

The quantity measured on them is the visible fraction $f = \|\Pi\tilde{\mu}\|^2/\|\tilde{\mu}\|^2$ of Theorem 2(iii), here the share of the measured error against experiment that lies in the auditable subspace. Pooled it is 0.6%, against those chance levels (Figure 2B1); per system 0.0065, 0.0017, 0.0086 and 0.0460. Dropping **cdk8** leaves the pooled figure at 0.6% (0.0060 against 0.0060) and moves it from 5.45% to 6.10% in the unweighted metric, so the pooled figure does not rest on it.

The per-system figures are less stable than the pooled one. Leaving out one edge at a time moves them by 1.9, 1.7 and 2.6 times on the first three and by 17 on **bace**, whose range spans a factor of 23.

Both sides of the ratio are measured in the variance-weighted metric the closure statistic lives in, and the margin over chance depends on that choice. It narrows to a factor of two on **bace** under isotropy in kcal/mol, and the pooled fraction rises to 5.45% in the unweighted metric, which recomputes the projector without variance weighting. Supplementary Table S2 gives all three conventions for all four systems with their own chance levels and their single-edge-deletion ranges; its weakest cell is **bace** isotropic under deletion, where f is smaller than its own chance level by a factor of 1.36.

The pre-registered criterion is met. It asked for a pooled degrees-of-freedom-adjusted gradient-model R^2 of at least 0.50, positive on three of the four systems; the pooled value is 0.980. These figures support a bound and not a point estimate: at its weakest the visible fraction sits a factor of two below its chance level, not the two orders of magnitude the pooled figure alone suggests.

The measurement is not an artefact of the fit, because it is a near-identity: per-ligand affinities make the experimental $\Delta\Delta G$ vector a gradient field, so $\Pi\varepsilon$ reduces to the closure residual and f is the closure statistic over the total squared standardized error. It survives the obvious contaminants, and agrees where a different label source can be run (Supporting Information). Per-edge error against experiment runs 1.06–1.45 kcal/mol at the median against reported standard errors of 0.14–0.40: these networks are wrong by several standard errors while closing their cycles, in the one direction closure cannot look.

The design does not license attribution. Experimental measurement error and the annotation join are per-ligand, hence gradient fields too, so what is established is that the dominant part of the error is node-consistent, not that it is force-field bias.

The same measurement over the wider curated set. The curated cross-check named above is not confined to the four systems the test selected, and it is the second block of the same panel. That source shares fourteen system names with the replicate set, and eight of them keep a cycle once the two edge sets are matched by structure rather than by name. Pooled over those eight, on 205 edges and 31 residual degrees of freedom, the visible fraction is 0.00052, against chance levels of 0.151 whitened and 0.0137 isotropic (Figure 2B2; Supplementary Figure S6 and Table S3; `make figGround`). The systems the closure test does not flag show the effect at least as strongly as the ones it does, 0.00042 over six against 0.00071 over two, which turns the measurement from a statement about test-selected networks into a statement about the benchmark.

The expanded measurement neither replaces the four-system one nor corrects it, which is why Figure 2B keeps the two as separate blocks rather than as one series. `bace` is absent from the curated set entirely, `p38`’s matched sub-network is acyclic and contributes to neither side, the two run on different edge sets, and the expanded set carries 31 residual degrees of freedom against the original’s 51: the two are measurements of different things on different labels, and both are reported. On genuinely identical edges the cleaner labels give the *higher* f , by $4.4\times$ on `cdk8` and $2.9\times$ on `hif2a`. That is the direction a per-ligand label error predicts and not its reverse, because such an error is a gradient field: the whitened projector annihilates it, so it lands wholly in the denominator of f and pushes the noisier source’s reading down.

One degree of freedom in the protocol was undeclared; the naming finding of Section 6 settles it. A ligand name is not an identifier here, so the figures quoted throughout key nodes by resolved structure, which is the less favourable reading: merging two names that denote one molecule knits two components into one, and the cycles that creates can only raise the auditable share. `ptp1b` is the only system of the eight it touches, going from $f = 0.0023$ to 0.0159, its worst single-edge-deletion isotropic margin from $10.3\times$ to $2.0\times$, and the pooled figure from 0.00046 to the 0.00052 above. Neither key is safe in general, and the Supporting Information shows where this one fails: a 2D structure merges poses that a release deliberately names apart. Both readings are reported.

Why the label source can only shrink the fraction. The label source can move f in one direction only, which is what makes it a bound. Those per-ligand errors reach an edge as a difference of node offsets, exactly the gradient fields part (i) describes, and the whitened projector annihilates them. Inject a per-ligand field at the mixed-laboratory reproducibility public affinity data show, 0.68 $\log_{10}$ units per ligand [13]. Over 800 seeded draws it moves the numerator of f by at most

1.96×10^{-11} of itself, against a norm of order one, while the denominator grows by 1.6 to 2.2 times at the median (Supplementary Figure S7; `make_figNoise`). Label noise therefore contributes nothing to the numerator and all of it to the denominator, so a correction for it can only raise f , and the measured 0.6% is a floor on the share attributable to the calculation’s own error and not an estimate of it.

The room that leaves is not uniform across systems, and the pooled crossing is not the reading to quote alone. What a correction could deliver is the reverse accounting: removing the share a literature-scale per-ligand noise implies would shrink the denominator by $4.13\times$ pooled, once each system is held to its own arithmetic.

The two metrics accordingly give different verdicts, and both are reported. The whitened reading, the metric f is defined in, survives every level tried on every system with one cell excepted. That cell is **bace** under its least favourable single-edge deletion, where the shrinkage needed and the shrinkage supplied coincide to within one per cent, which is a tie rather than a margin. The isotropic reading does not survive on **p38** and **bace**, the two weakest cells of that convention. The per-system arithmetic behind both verdicts is tabulated in the Supporting Information.

One level is ruled out by its own arithmetic. At the full mixed-laboratory level the implied label-noise share exceeds one on **cdk8**, **p38** and **bace**, asserting more label variance than those networks’ entire measured error against experiment contains. That level is an upper bound and not an estimate for a join that takes one assay per system and never mixes IC_{50} with K_i .

Grounding against measured affinity. On **eg5**, whose ligands carry ChEMBL identifiers, the network $\Delta\Delta G$ from the same generalized-least-squares fit agrees with real KIF11 experimental affinity [32] at a mean unsigned error (MUE) of 0.79 kcal/mol (95% bootstrap CI [0.59, 0.97], conditional on the fitted network; Pearson $R = 0.65$ [0.30, 0.83], $p \approx 2 \times 10^{-4}$, $n = 28$), and its clean cycle-closure reduced $\chi^2 = 0.64$ is consistent. That is not evidence that a clean system is accurate, and the five grounded systems cannot settle the question in either direction.

The ordering by mean unsigned error is **eg5** 0.79 = **p38** 0.79 < **bace** 0.88 < **cdk8** 0.91 < **hif2a** 1.47, so the one unflagged system ties the best flagged one and is below the other three. That direction favours the flag, but it rests on a single unflagged system against four, and the grounding is not matched: **eg5** comes from a single named ChEMBL assay while the four flagged systems come from a target-wide connectivity search under the coverage rule. No separation claim follows from it in either direction.

Acting on the flag does not improve accuracy on **eg5** (internal $|z|$ against error-against-experiment, $r = +0.13$), which is what the scope predicts: on a clean system the residual error against experiment is dominated by node-consistent force-field bias, and cycle closure cannot see that. The complementary test, that repairing a flagged system improves its accuracy, needs experimental affinities for the flagged systems; these are not directly tabulated in the replicate set but are recoverable by per-paper ChEMBL structure-search. Of the six flagged systems, four clear a 60% public-ChEMBL coverage gate (**cdk8**, **hif2a**, **p38**, **bace**; connectivity-based structure matching, one assay per system, IC_{50} before K_i , never mixed); their baseline network MUE against this grounded experimental affinity is 0.79–1.47 kcal/mol, typical-magnitude FEP error, which confirms the grounding itself is sound.

Removing the flagged edges. On these four, guided removal of the flagged high- $|z|$ edges (the same repair order as Figure 3A, fixing a removal count $K = 1\text{--}3$ per system) does not reduce MUE-against-experiment more than an equal-size random removal does (Supplementary Figure S10; per-system one-sided permutation $p = 0.905, 0.136, 0.880, 0.071$; combined Stouffer $p = 0.48$, sign test $p = 0.69$). Two systems improve slightly over their random-null mean and two worsen slightly, with

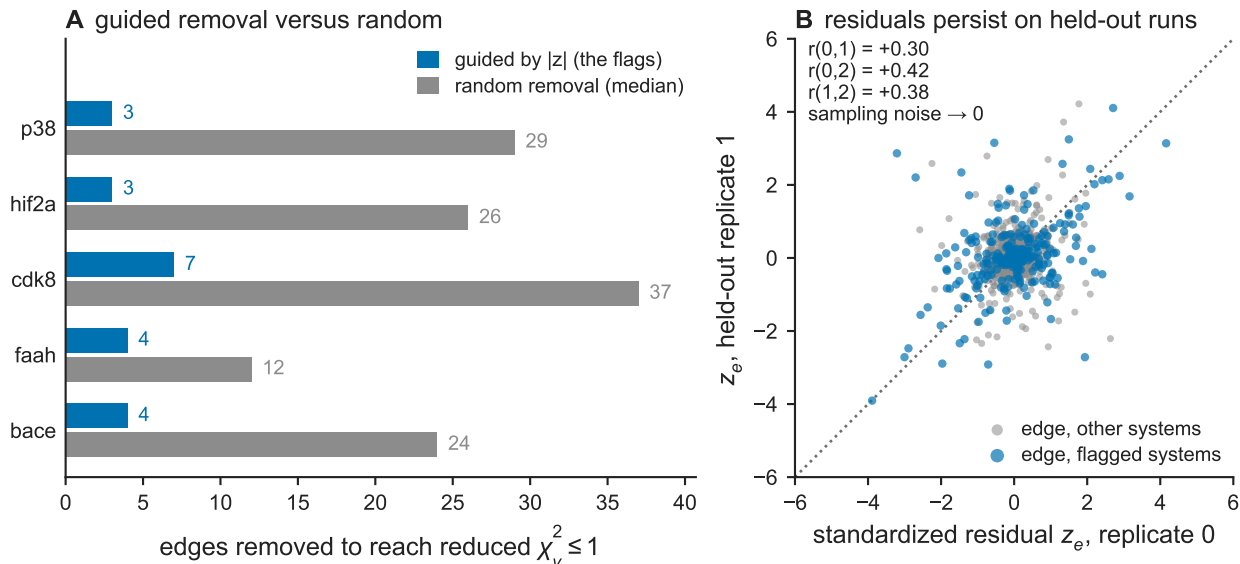


Figure 3: **Edge-removal repair test and cross-replicate residual reproduction.** (A) Number of edges removed, in descending order of $|z|$, before reduced $\chi^2 \leq 1$ is restored: guided removal of the flagged high- $|z|$ edges versus random removal, per system. (B) Standardized residual z_e from a replicate-0 network fit versus z_e from a replicate-1 fit, for edges in flagged and in non-flagged systems, with the $y = x$ line and the Pearson correlation between each pair of replicates.

no system individually significant. The frozen criterion is conjunctive, requiring Stouffer $p < 0.05$ and a guided ΔMUE , defined throughout as $\text{MUE}(\text{full}) - \text{MUE}(\text{after})$ so that an improvement is positive, above the 95th percentile of its own random null in a majority of grounded systems; neither conjunct is met. The remaining two flagged systems (**brd4**, **faah**) did not clear the coverage gate and are not included.

That null is what Theorem 2 predicts and what the paragraph above measures. A removal rule reads the visible part of the error; the visible part carries under one per cent of what is wrong with these networks, and by part (ii) it is the part that biases the fitted potentials least. A second pre-registered rule tests whether the ranking was simply the wrong one, and fails to find that it was; a second failure to reject is weaker evidence than a rejection would be. Part (ii) suggests ranking edges by influence rather than by residual. The statistic used is $|z_e|/\sqrt{(1-h_e)/h_e}$ with $z_e = r_e/\sqrt{V_e^{\text{rep}}}$ the raw standardized residual, which is the classical influence form scaled by $\sqrt{h_e}$ and not that form itself; edges with h_e below 10^{-9} are treated as bridges and given influence zero. Writing it in the unscaled form changes the removal set on two of the four systems, and that variant is reported with the outcome below.

At the same removal counts the influence rule performs worse than the rule it was meant to replace, against the same conjunctive criterion. It is better on one system of four, at a pre-registered sign test of $p = 0.94$ against the residual rule’s 0.69 (per-system ΔMUE -0.055 , -0.099 , -0.612 and $+0.140$ kcal/mol, at one-sided permutation $p = 0.937$, 0.819 , 1.000 and 0.023 ; Figure 2C). On four systems that test attains only $\{1.0000, 0.9375, 0.6875, 0.3125, 0.0625\}$, so neither value could reach its own level and the two do not form an ordering, one system flipping separating them.

The pre-registration’s other combination, a one-sided Stouffer, returns no usable number here. One system’s permutation p saturates at the resolution of its own grid and Stouffer maps that to an infinite z ; clipping to the grid’s range gives 0.96, and the residual rule, which has no saturated

input, stands at 0.48.

Neither rule beats random removal, and on one system the influence rule does measurable harm. On **p38** its guided ΔMUE of -0.612 kcal/mol falls below every one of the 1000 random draws (minimum -0.416 , no ties), so the opposite tail is at most $1/1001$. That tail is post hoc, the pre-registration having asked whether the rule helps; it is one system of four, about 0.004 under a Bonferroni allowance; and the magnitude is convention-dependent, -0.612 under greedy removal that refits after each edge and -0.232 when the statistic is applied once to the original fit. The sign survives both conventions, and survives the strict influence form, which changes the removal set on two of the four systems but not here. The two edges removed carry $h_e = 0.099$ and 0.388 , so the harm is not the degenerate branch where a near-bridge is divided by its own vanishing leverage.

The other tail points the other way and is reported alongside it. On **bace** the same rule improves the error against experiment by 0.140 kcal/mol at a one-sided $p = 0.023$, which the conjunctive criterion does not let the summary report, and which is exactly as post hoc and as single-system as the harm. Neither tail clears the pre-registered bar, and both are reported.

The rule fails overall for the reason measured above: the influence form divides a residual by a small curl-leverage to recover the error of a single anomalous edge, and a smooth per-ligand field is not concentrated on any edge, so the division returns amplified noise.

A flag remains a system-level signal that a protocol has produced results inconsistent beyond sampling, and that signal reproduces across replicates in its per-edge residuals and in its ranking of systems, though not in the list of names it returns. It is not a repair target: deleting the edges it names cannot recover accuracy against experiment, which is set by a component the audit is blind to by construction.

Reaching that component needs information from outside the network, and it has to be information the network does not already contain. An absolute binding free energy calculation carries exactly the force-field component of the per-ligand systematic error part (i) is about, so it does not resolve that component at all. The mapping, soft-core, dummy-atom, schedule and restraint contributions are protocol-specific and would not transfer. A measured anchor resolves its dimension only to its own precision. By part (iv) the requirement does not soften when the anchors are unbiased: one anchor per component fixes only an arbitrary offset, and every dimension after that costs one more measured ligand.

5 The per-edge error bar as a null hypothesis

The test of Section 6 needs a per-edge variance it can treat as known. Two quantities are involved, and they are kept apart throughout: the closed form for a single BAR window, which the replicate benchmark cannot exercise because it releases no work samples, and the bar the test actually carries, which is the multistate standard error that benchmark ships, composed in quadrature from the two legs and validated against independent repeats. Everything below consumes the variance as a number or as a fixed graph weight.

Theorem 3 (Sampling variance via the sandwich). *For one alchemical edge with n_f forward and n_r reverse works, let p be the logistic BAR weights at the root $\widehat{\Delta F}$ of Bennett’s estimating equation $S(\Delta F) = 0$ and $I = \sum p(1 - p)$ its curvature, the work-distribution overlap. Then $\widehat{\Delta F}$ is an M -estimator, so its asymptotic sampling variance is the sandwich $\text{Var}(\widehat{\Delta F}) = A^{-1}BA^{-1} = B/I^2$, with $A = I$ and $B = n_f \text{Var}_f[p] + n_r \text{Var}_r[p]$ [11]. Here $A \neq B$ arises from BAR’s stratified two-sample design and not from model misspecification: the information equality $A = B$, which would give $1/I$, holds for a single-sample MLE and fails for a two-sample one. Evaluated at the root, with works drawn from the correct pair of ensembles, $B/I^2 = 1/I - (1/n_f + 1/n_r)$ identically: the sandwich*

is not an approximation to the Bennett/Shirts–Chodera (MBAR) uncertainty [4, 25] that `pymbar` reports but the same quantity, and $-(1/n_f + 1/n_r)$ is exactly the $O(1/n)$ gap between it and the naive plug-in $1/I$. The two sample plug-ins still differ through the moments they substitute, by a paired discrepancy falling monotonically and by more than an order of magnitude from $n=40$ to $n=320$, the range over which it exceeds Monte-Carlo noise at a modest replicate budget.

Remark (scope of “calibrated”). B/I^2 is the variance of the *sampling* error of $\widehat{\Delta F}$, conditional on the works being drawn from the correct alchemical ensembles. It says nothing about systematic force-field bias, which is outside B/I^2 ; a separate epistemic correction term is attached where the downstream uses of the uncertainty are audited (Section 8). It is also non-negative by construction, where the plug-in expression can return a negative value at very high overlap with tiny samples.

Figure 4 checks the closed form on real work samples, the 19 adjacent- λ BAR windows of the complex leg of the `alchemtest` BACE1 relative-binding calculation. The sandwich over decorrelated-bootstrap truth is 1.00 [0.96, 1.07], against 2–12 $\times$ for naive $1/I$ on the same reference, every window sitting at overlap ≥ 0.84 (per-window values, Supplementary Table S1). That reference reuses the same work samples, so the figure establishes agreement with a decorrelated bootstrap on real molecular dynamics in the high-overlap regime this schedule occupies, and not a ligand-to-ligand binding quantity; and since the replicate benchmark releases no work samples, the work-sample route to B/I^2 is exercised on these 19 windows alone. Both the sandwich inputs and the bootstrap truth are subsampled to the statistical inefficiency g [25] before B is formed, so Figure 4 uses effective n (Supplementary Figure S1).

The low-overlap behaviour rests instead on a controlled Monte-Carlo overlap sweep, in which the closed form tracks MBAR and Monte-Carlo truth across the range at a reported/true se of ≈ 1 and at no Monte-Carlo cost, bare $1/I$ overstates by 1.08–2.65 $\times$, worst at high overlap, entering as the information-equality plug-in the sandwich corrects and not as a baseline anyone reports, and a learned mean–variance head given the same overlap information is overconfident at both label budgets tested; at extreme low overlap ($n=20$) the bias-corrected plug-in itself runs $\sim 10\%$ below MBAR, both ≈ 1 . That last result is a property of the head’s per-edge parameterisation and not a limit on learning the quantity: pooling the same single labels by overlap recovers the sandwich to $\sim 6\%$, and a null miscalibrated by that much leaves the test of Section 6 intact. The sweep, and the two restrictions that travel with the learned head, are Supplementary Figure S2.

The reported bar against independent replicates. Figure 5 validates the bar the test actually uses, on samples it does not reuse. Each edge’s reported bar is composed in quadrature from the complex-leg and solvent-leg MBAR uncertainties shipped with the benchmark, $se = \sqrt{se_{\text{complex}}^2 + se_{\text{solvent}}^2}$. Each leg term is a multistate MBAR uncertainty over that leg’s λ windows and not the two-state BAR quantity of Theorem 3, so what is validated here is the bar as a practitioner receives it and not the two-state identity itself.

Across 1145 real binding edges from the OpenFE IndustryBenchmarks2024 [2] (3 independent replicates each), the reported MBAR se is calibrated-to-conservative against the across-replicate truth: the replicate ratio is 1.41 [1.26, 1.62], and the bar is conservative on 72% of edges and on 28 of the 34 systems carrying at least eight edges. Both sides of that ratio pool in quadrature, so the $n=3$ replicate term is unbiased at the variance level and needs no small-sample correction. The three replicates are independent stochastic repeats of the sampling process from identical starting coordinates (OpenFE `protocol_repeats` [2]), so the across-replicate SD reflects sampling and convergence noise alone and is a *lower bound* on full reproducibility, excluding pose and system-preparation variance. The reported se does not systematically under-state real reproducibility.

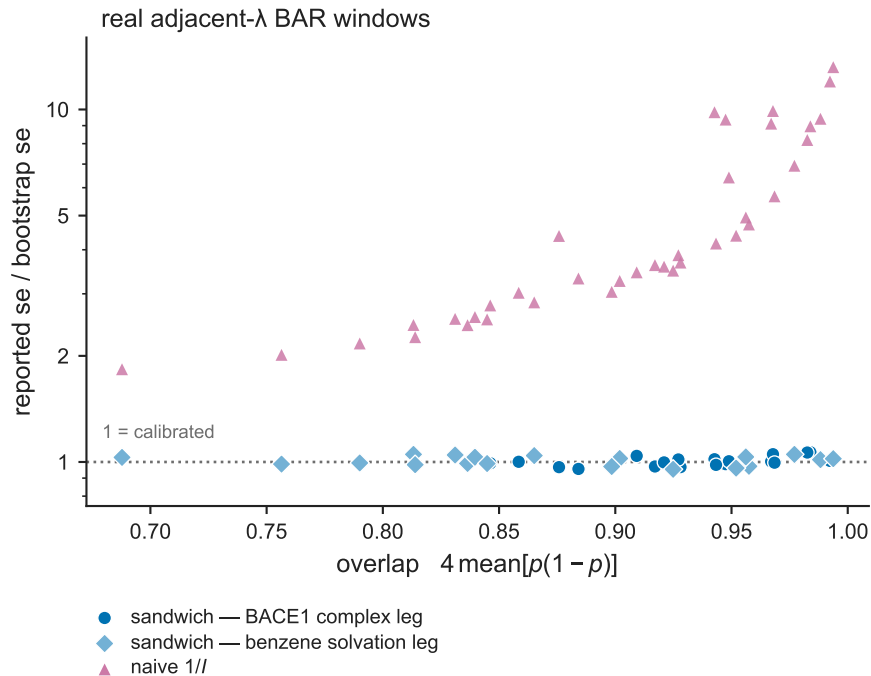


Figure 4: **Sandwich variance against a decorrelated bootstrap on real BAR windows.** Reported standard error divided by the decorrelated-bootstrap standard error, versus work-distribution overlap, for the sandwich B/I^2 on the 19 adjacent- λ BAR windows of the complex leg of the `alchemtest` BACE1 relative-binding calculation (filled circles) and on the 19 windows of the benzene solvation legs (filled diamonds), and for the naive information-equality plug-in $1/I$ on the same windows (triangles). Each point is one two-state BAR estimate between neighbouring λ values, and every window sits at overlap ≥ 0.84 . The controlled low-overlap sweep and the learned-variance baselines are Supplementary Figure S2.

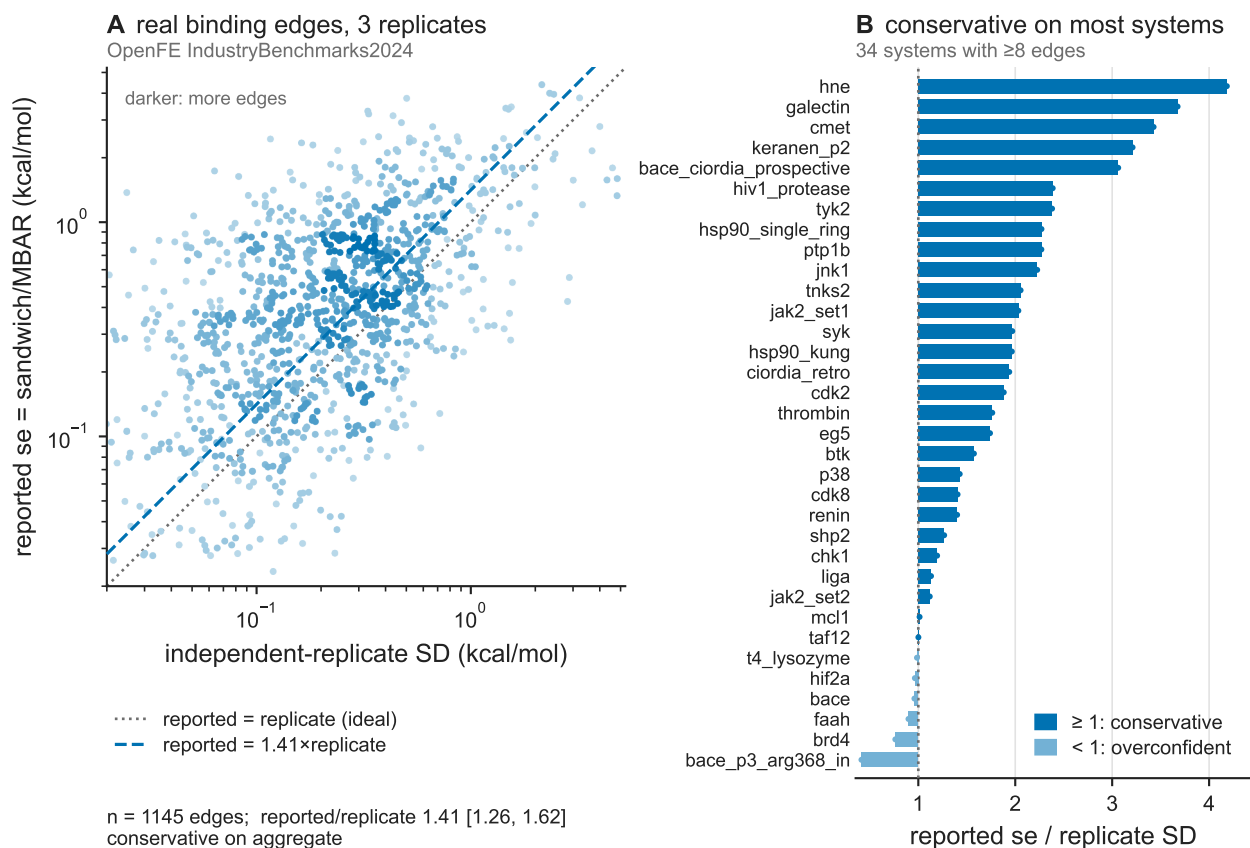


Figure 5: **Independent-replicate validation on real binding edges.** Reported per-edge standard error versus the across-replicate empirical standard deviation, on 1145 OpenFE IndustryBenchmarks2024 [2] edges (3 independent replicates each). (A) Reported standard error versus across-replicate standard deviation, with the $y=x$ line. (B) Per-system pooled ratio of reported standard error to across-replicate standard deviation, for the 34 benchmark systems carrying at least eight edges.

That conclusion disagrees with the closest measurement in the literature. Wade et al. [27], comparing the analytic MBAR uncertainty against the spread over five independent replicas on 54 protein-ligand transformations across five targets in two engines, report that MBAR consistently under-estimates the error, by up to 0.9 kcal/mol. Two differences bear on the disagreement, in opposite directions. Their replicas resample the full ensemble, whereas the OpenFE repeats share starting coordinates, so the denominator used here is a lower bound on run-to-run variability and the ratio an upper bound on conservatism. Their own functional, applied to these edges, sharpens the disagreement instead of resolving it: the analytic standard error falls below the replicate spread on 0.279 [0.224, 0.332] of edges, against the 0.368 correctly scaled bars produce at three replicates, and the interval excludes that value. Settling the question needs replicates that resample preparation and pose, which this benchmark does not provide, and its eight multi-release systems supply no usable substitute (Section 6).

6 The test, and what it inherits from the bound

The calibrated closure test, called the quality-control (QC) test in this article’s figures and the calibrated test below, makes the bound operational: the residuals of the same projector whose diagonal Theorem 1 reads as an observability certificate, weighted by the bar of Section 5, are a statistic with a stated null. It separates the reproducible part of a perturbation network’s edge-level inconsistency from its sampling noise. Candidate physical sources of a reproducible part include non-convergence, hysteresis, and charge, water or protonation artifacts, but this design does not identify which of them is present, and it does not separate them from a fixed consequence of the one starting structure every repeat begins from.

The calibration underwriting the null is validated against seed and velocity replicate spread, a lower bound on full run-to-run reproducibility, since it excludes pose and system-preparation variance. That restriction bounds what the detection claims below can mean: an inconsistency that survives repeats sharing their starting coordinates is established as more than sampling noise, and not as physical error.

The test performs detection rather than weighting of a fixed estimate. The Gauss–Markov ceiling that makes the calibrated variance second-order for ranking, in a connected graph the GLS contrast mean being unbiased for any positive weights, does not bind a goodness-of-fit test, whose statistic depends on the weights directly.

A calibrated cycle-closure test. The fit of Section 3 gives residuals r_e and the goodness-of-fit statistic $X^2 = \sum_e r_e^2 / V_e^{\text{rep}} \sim \chi^2(\text{dof})$, with dof the number of independent cycles (E minus the incidence rank). Under correctly-sampled physics $\mathbb{E}[X^2/\text{dof}] \approx 1$; here it runs below one, at a median reduced χ^2 of 0.34, qualitatively consistent with the conservative bars of Figure 5. A reduced χ^2 well above one signals edge-level inconsistency in excess of sampling noise, and the test is correctly blind to node-consistent force-field bias, which cancels around a loop.

Two variances have to be told apart from here on. The variance a practitioner is handed, V_e^{rep} , is what the fit uses as its weight; the variance the edge actually has is V_e^{true} . They are related by $V_e^{\text{rep}} = c_e^2 V_e^{\text{true}}$, so the per-edge calibration ratio c_e is the reported standard error over the true one: $c_e > 1$ is a reported bar that is too wide (conservative) and $c_e < 1$ one that is too tight (overconfident). On this benchmark c_e is measured as the reported standard error over the across-replicate standard deviation.

Three summaries of that distribution appear in this article, and they are different functionals of it rather than competing estimates of one number: the replicate ratio 1.41 (Figure 5), the closure-

weighted mean 0.85 whose root is the calibrated scale 0.92, and the median reduced χ^2 . They disagree because the per-edge ratio is heavy-tailed, spanning a factor of 128 between its first and ninety-ninth percentiles while the summaries occupy a $1.57\times$ window. No single scalar conservatism factor is identified by these data, and the apparent disagreement is an artefact of summarising one heavy-tailed distribution three ways. The definitions, the distribution with the summaries marked on it, how much of its spread three replicates guarantee even when every bar is right, and a check that residual correlation among edges sharing a ligand endpoint leaves the flag set unchanged, are all Supplementary Figure S13 (`make figCal`).

Four things here are established: network maximum-likelihood estimation of $\Delta\Delta G$ networks [30], cycle-closure hysteresis as a diagnostic [5, 28], weighted cycle closure [18], and the constrained variational solution of a whole network under loop closure [6]. What is new is the *calibrated* null, a per-edge aleatoric variance measured against seed and velocity repeats instead of assumed, which gives the test per-edge adaptive thresholds and complements the fixed hysteresis cutoffs in standard use. Those repeats share one pose and one system preparation, so the variance they validate excludes the component the alternative hypothesis is built from, no controlled false-positive rate follows from them, and nothing on this benchmark flags once the bars are widened to cover that component.

The calibrated null comes together with the cross-replicate reproduction check of Figure 3B and the estimation–detection conservation law of Theorem 1, which fixes which edges are auditable at all. What is shared with the works above is the weighted least-squares treatment of the cycle graph itself.

On the same 1143 edges a pre-registered head-to-head scored the calibrated test against the standard fixed 1.0 kcal/mol hysteresis cutoff and against a fixed-se χ^2 test, by area under the receiver-operating-characteristic curve; the criterion required beating both, and the verdict was a tie (Supplementary Figure S11). No further reading attaches to the ordering inside that tie, because the only label these data supply divides each residual by the same per-edge variance the calibrated rule uses as its weight: the ordering is monotone in proximity to that variance and not in detection skill, and the comparison ranks nothing.

The two rules are also not independent. On replicate 0 the three flag sets are strictly nested, the calibrated test flagging 6 systems, the fixed-se test 11 and the 1.0 kcal/mol cutoff 26, so the calibrated test catches nothing the cutoff misses and instead removes 20 of its flags.

The three-way chain does not survive the other repeats intact: on replicate 1 the calibrated test flags 6, the fixed-se test 13 and the cutoff 24, and two systems break it, `ciordia_retro` flagged by the calibrated test but not by the fixed-se test and `mup1` flagged by that test but not by the cutoff. On replicate 2 the chain holds again at 3, 11 and 24. The containment any claim here rests on is the two-set one, between the calibrated test and the cutoff, and it holds on all three replicates.

Whether removing those 20 flags is an improvement is what the contaminated label cannot say, and adjudicating them needs new sampling or new experiment, which public retrospective data cannot supply. The calibrated test’s advantage over a fixed cutoff is accordingly structural and is not demonstrated here: per-edge adaptive thresholds, a null measured against seed and velocity repeats instead of assumed, and the observability map of Theorem 1. That measurement is not a controlled error rate, since the repeats hold pose and preparation fixed.

The quoted level is nominal and not verified. The null is conservative in the bulk, at the median reduced χ^2 above, but a conservative bulk and an anti-conservative tail can coexist, and the median of the statistic does not bound the realized false-positive rate. The case for the flags therefore rests on the distribution-free cross-replicate reproduction check below, which is independent of the χ^2 scaling and assumes no Gaussian null.

Two robustness questions are settled in the Supporting Information rather than assumed away. Replacing Benjamini–Hochberg by Benjamini–Yekutieli [3], which needs no dependence structure at

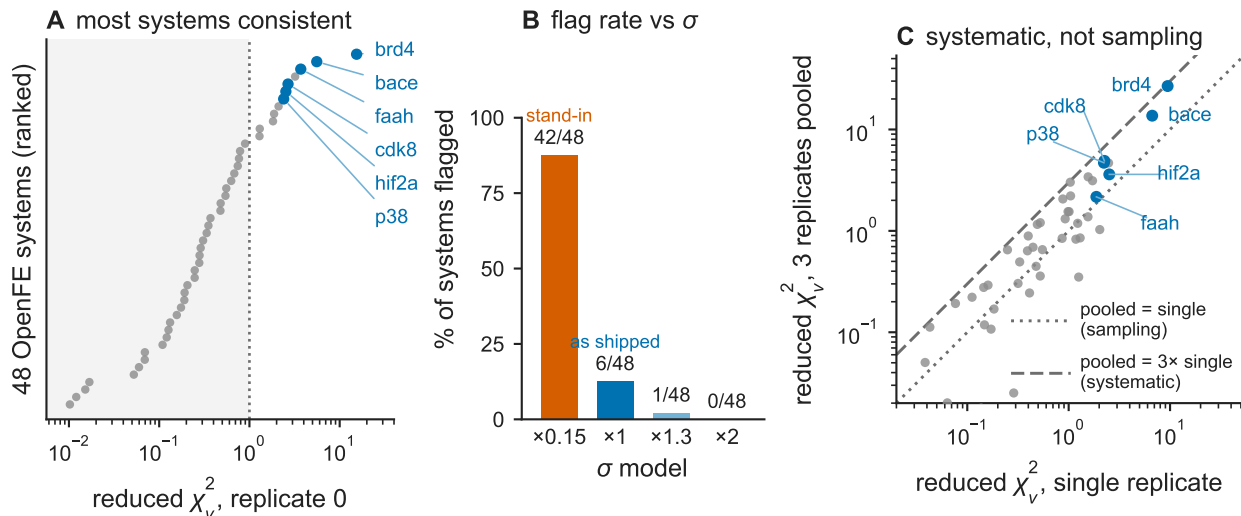


Figure 6: **Calibrated cycle-closure quality control.** (A) Per-system reduced χ^2 from a generalized-least-squares cycle-closure fit, ranked, with shading below $\chi^2_\nu = 1$ and the flagged systems marked. (B) Percentage of systems flagged under four σ models, in plotted order: a uniform $\times 0.15$ shrink, the reported bar as shipped, a $\times 1.3$ inflation and a $\times 2$ inflation. The tick marked $\times 1$ is the shipped bar and not the calibrated scale. (C) Reduced χ^2 from a single replicate against the value from the three replicates pooled, with reference lines at pooled/single = 1 and = 3.

all, costs a factor $H_{48} = 4.46$ in the threshold and returns the same six systems on replicate 0. And four of the six flagged systems are themselves locally overconfident against their own across-replicate spread, so a flag could partly reflect that system’s own too-tight bars; retesting each against the width its own replicates measure keeps all six (Supplementary Table S9).

The test on the replicate benchmark. The OpenFE IndustryBenchmarks2024 release [2] carries 1145 edges over 49 systems in three replicates, of which the 48 systems with at least one independent cycle, holding 1143 edges, admit the test. On those, the median reduced χ^2 is 0.34. A Benjamini–Hochberg false-discovery-rate (FDR) test at 0.05 applied to the first replicate flags six chemically-sensible systems (**brd4**, **bace**, **faah**, **cdk8**, **hif2a**, **p38**; Figure 6A; **brd4**’s data set is a buried-water set, **bace** is protonation-sensitive). That particular set of six is a single-replicate draw and should not be read as a census; Section 6.1 measures how much of it survives the other two replicates.

One of the six has an identifiable cause. The benchmark labels a system by target, and eight of its names union edges from more than one separately released data set. Where those releases share ligands the union carries cycles present in neither release alone, and the cycle spaces sit orthogonally inside the union’s, so the closure statistic splits exactly. On **cdk8** the split is decisive: the union gives $\chi^2 = 77.7$ on 29 degrees of freedom, of which the two releases carry 31.1 on 23 between them, leaving 46.6 on the six cycles that span them, a reduced χ^2 of 7.8 at $p = 2.3 \times 10^{-8}$. Neither release is inconsistent on its own: **merck_cdk8** (51 edges, 20 degrees of freedom) gives $p = 0.218$, 0.865 and 0.189 on the three replicates, and **misc_cdk8** (12 edges, 3 degrees of freedom) gives 0.091, 0.992 and 0.985, the first being the smallest of the six and still well short of any threshold used here. The spanning cycles exist because the two releases share seven ligand *names*, and all seven denote chemically different molecules in the two releases, name 13 being a benzyl-indazole carboxamide in

one and a chlorophenoxy-thienopyridine carboxamide in the other. Every spanning cycle therefore asserts that two distinct ligands have the same free energy, and resolving each edge within its own release removes all six on every replicate, taking the degrees of freedom from 29 to the within-release total of 23. The 46.6 is a bookkeeping collision in the benchmark’s ligand naming and not a disagreement between two preparations. `cdk8` is flagged for that collision, and the flag is retained in the published set with its cause named, because a system whose network asserts a false identity is what an inconsistency test should reject.

The hazard belongs to the analysis and not to the released data. Each release is internally consistent and each names its own ligands unambiguously; what fails is the join, and it fails for anyone who unions two releases of a target on ligand names without checking that the names denote the same molecules. That is a common enough way to assemble a benchmark network that it is worth saying plainly: match on structure, not on name.

A closure statistic reads the network it is given, and an inconsistency produced by the bookkeeping that assembled that network enters as the same cyclic residual as one produced by the simulations. No other system carries across-release cycles that matter. Two shared ligands are needed before a union carries a cycle that spans its releases, and in six of the eight the releases share at most one. There the union is disconnected or joined by a tree, and no such cycle exists at any magnitude. `tyk2`, whose two shared names do denote the same molecules, gives one spanning cycle that is unremarkable ($p = 0.67$). The single cross-release comparison that joins matching ligands shows no excess.

What the benchmark supplies on between-preparation variance is therefore a negative result rather than a number. With `cdk8`’s six spanning cycles resolved away as a naming collision, the only genuine cross-release join anywhere in the benchmark is `tyk2`’s single cycle. It never departs from its null, at $p = 0.67$, 0.40 and 0.84 on the three replicates, and its across-release block never rises above 0.64 of the within-release one (Supplementary Figure S8; `make figRelease`). The public data contain no usable estimate of what a second preparation of the same target adds, which bounds what can be said about the pose and preparation component throughout this article rather than supplying it. Controlled work does bound it from outside. Starting one alchemical edge from two kinetically distinct conformations of the same protein shifts its free energy by as much as 4.74 kcal/mol under a default protocol. The root-mean-square inconsistency is 4 kcal/mol, and falls to 0.57 only once the runs are extended to 55 ns; the effect is put at 2–5 in general [19]. Independently, careful preparation of protein and ligand structures is what brings a leading workflow to the reproducibility of the experiments it is scored against [23]. The mechanism is kinetic trapping rather than thermal averaging, so a wrong starting state persists as a fixed offset for the whole run. Repeats that all begin from that state cannot price what it costs.

The grouping is therefore a choice the analysis makes, and the flag set is not invariant to it. Re-running the same replicate-0 test on the release’s own blocks gives 56 testable systems and again six flags, but a different six: `cdk8` leaves and `jacs_mc11` enters. The grouping by target name was not pre-registered, and the Benjamini–Hochberg family size depends on it. The published list is the by-target one throughout. Supplementary Tables S6 and S7 give the per-replicate degrees of freedom, reduced χ^2 and adjusted q -value behind every flag.

The test’s selectivity is entirely a function of calibration (Figure 6B). A uniform $\times 0.15$ shrink standing in for the learned head’s central overconfidence (Supplementary Figure S2) flags 88% of systems, 42 of 48, which leaves almost nothing unflagged and so no longer separates anything. The reported bar as shipped flags 12%, 6 of 48; a $\times 1.3$ inflation one system; a $\times 2$ inflation none. These are flagged fractions and not false-positive rates: which of the flags are false is not known here (Section 8).

The count is not separable from the level, and the shipped operating point is the conservative

one. The scale at which the reported bars are correct is the calibrated scale, which follows from the closure-weighted mean, the functional that governs the statistic, and not from a pooled ratio; at that mean the bars are wide by $1.09\times$. Section 6.2 puts the closure-weighted mean in $[0.63, 1.08]$, and the calibrated scale accordingly in $[0.79, 1.04]$. Correcting every reported bar to a scale in that interval and re-running the test on the real data gives six flags at 1.04 and at 1.00, seven at 0.92 and nine at 0.79 (`make figSelf`).

A pure-null simulation puts the realized probability that the test flags anything when nothing is wrong at essentially zero when the bars are as wide as they are measured to be, and at 0.100 when they are slightly tighter than the truth. The exposure that matters is therefore the anti-conservative end, where a bar a little too tight doubles the nominal rate. That simulation verifies the implementation rather than the real-data operating point, which is again why the level above is nominal. The scales involved are reciprocal and easy to confuse, so the grid, its exact binomial intervals and what four hundred draws can and cannot separate are set out in the Supporting Information.

Sensitivity is fragile in the other direction. At $1.3\times$ one system of 48 flags, and at $1.5\times$ and $2\times$ none do, so nothing on this benchmark survives a bar widened to cover pose and preparation variance. Widening changes the null being tested rather than moving the same test along it, so what is fragile there is sensitivity and not validity.

The uniform shrink is a stress test and not a realistic head, and the comparison is a counterfactual in a further sense: no deployed or published pipeline is known to feed a learned per-edge σ into a network consistency check. The head’s own measured, overlap-dependent profile applied per edge flags 42 of 48 as well, so the collapse is not an artefact of uniformity. The demonstration is accordingly scoped to heads as overconfident as the measured one, whose reported errors run $5\text{--}11\times$ too small (Supporting Information).

The three independent replicates confirm the flagged inconsistency is reproducible and not sampling. Pooling the replicates shrinks the bars ($se \rightarrow se/\sqrt{3}$) yet increases the flagged systems’ reduced χ^2 toward the $3\times$ variance-components ceiling (Figure 6C), which occurs only for a component that repeats (single-replicate $\chi_\nu^2 = 1 + \sigma_{\text{sys}}^2/V_{\text{samp}}$ vs pooled $1 + 3\sigma_{\text{sys}}^2/V_{\text{samp}}$, with σ_{sys}^2 the reproducible variance component and V_{samp} the sampling one; the ratio $\rightarrow 3$ iff reproducible-dominated). The argument fixes the component’s variance and not its origin: it is invariant to whether σ_{sys}^2 is physical error or a fixed consequence of the one starting structure the three repeats share, both being constant across repeats.

Cross-replicate reproduction of the flagged inconsistency. Greedily removing the flagged (highest- $|z|$) edges restores consistency (reduced $\chi^2 \leq 1$) in 3–7 edges, versus 12–37 removed at random (Figure 3A). That contrast illustrates the repair procedure and is not evidence, because it cannot fail: the statistic being driven down is $X^2 = \sum_e r_e^2/V_e^{\text{rep}}$ and the removal order is descending z_e^2 , that sum’s own per-edge summand, so a network with no localized systematic error at all reproduces the same contrast.

The evidential check is instead out-of-sample. Fitting each replicate’s network separately, the per-edge standardized residuals correlate across independent replicates ($r = +0.30$ to $+0.42$, $n = 1143$, each interval excluding zero under a bootstrap over systems; Figure 3B), where sampling noise alone would give zero. What the flags identify is therefore reproducible across runs and not a property of one draw. The correlation carries the same restriction as the pooling argument: any per-edge quantity held fixed across the three repeats produces it, the shared starting structure is such a quantity, and what is established is reproducibility and not a physical origin.

Summary. The calibrated bar supplies, at no label cost, the null that turns cycle closure into a test separating reproducible from sampling inconsistency, and a bar as overconfident as the learned σ measured here destroys it. Theorem 2 bounds what that test, or any other of its kind, can be asked for, which is why its flags are not a repair list.

6.1 What survives a second replicate

The benchmark ships three independent repeats, so the flag set can be recomputed instead of assumed, and recomputing it shows that the set does not reproduce.

Two conventions bound how the union below should be read. It is descriptive and carries no controlled error rate. Three tests each holding a nominal 0.05 would put the chance of at least one false flag among them at $1 - 0.95^3 = 0.143$ if they were independent, and these three are not; Benjamini–Hochberg in any case controls the false discovery rate rather than the probability of any false flag. The counts are also per system and not per target, the benchmark’s 49 systems carrying 41 protein labels, with **bace** contributing four and **hsp90** four; the six flagged systems are six different proteins, so that count at least is not inflated by system splitting.

The same pipeline at the same FDR level returns {**bace**, **brd4**, **cdk8**, **faah**, **hif2a**, **p38**} on the first replicate, {**bace**, **brd4**, **ciordia_retro**, **hif2a**, **p38**, **renin**} on the second and {**bace**, **cdk8**, **renin**} on the third. Eight systems are flagged at least once and one, **bace**, every time. Pairwise Jaccard overlap is 0.50, 0.29 and 0.29, against 0.056 for size-matched random draws, averaged over the three pairs. That instability is not a threshold artefact and persists from $\alpha = 0.01$ to $\alpha = 0.20$ (Supplementary Figure S12). Pooling the three replicates flags seven systems, the union minus **faah**.

The instability is not explained by how cyclic the networks are. Cycle counts on this benchmark are low: nine of the 48 systems carry exactly one independent cycle, twenty-three carry three or fewer, and the quartiles are 2, 5 and 12. A reduced χ^2 on one or two degrees of freedom is an unstable statistic, and those systems do vary more across repeats. But they are not the systems that move in and out of the flag set: seven of the eight ever flagged are well connected, and restricting the comparison to the best-connected third of the benchmark leaves the pairwise overlap no better than it is over all 48. The instability sits where the test has power, which makes it a property of the measurement rather than of thin networks.

Two properties do reproduce. The ranking of systems by closure inconsistency reproduces across replicates, at Spearman 0.649, 0.644 and 0.723 over all 48 systems. So does the dependence of selectivity on σ -calibration, which is the claim this section makes: across the three replicates the $\times 0.15$ shrink flags 42, 41 and 41 systems, the reported bar 6, 6 and 3, a $\times 1.3$ inflation 1, 1 and 1, and a $\times 2$ inflation none. A single replicate therefore supports a ranking and a calibrated operating point, and not a list of names.

Among the published six, **faah** is the weakest on every measure. Its reduced χ^2 crosses one in both directions across replicates (3.71, 0.45, 1.47), its standardized residuals barely reproduce between replicates ($r = -0.085$), and it is the one member the pooled fit drops: a flag a second replicate does not support, and not a member of the set on the footing of **bace**.

6.2 Out-of-sample behaviour

Two checks use one replicate to predict another and so cannot be circular. First, the per-edge mis-calibration measured from replicates two and three predicts the closure statistic of the replicate they exclude. With $V_e^{\text{rep}} = c_e^2 V_e^{\text{true}}$ as above, Theorem 1 makes the closure-weighted mean predictable from c_e alone, and that prediction correlates with the observed per-system χ_ν^2 at Spearman 0.648

($n = 48$). Both sides divide by the same reported variance, so the reference is not zero: under a simulation with no systematic error anywhere the same statistic returns 0.059 with a 95% upper limit of 0.364, and 0.648 exceeds every one of 200 draws (Supplementary Figure S14A).

The aggregate level agrees less well. The predicted closure-weighted mean is 0.85, with interval [0.63, 1.08]. The observed degree-of-freedom-weighted value is 1.32, with interval [0.81, 1.80]. The two intervals overlap, and the prediction does not cover the observation.

The scope of that agreement is narrow. The prediction is built entirely from measured miscalibration, so a rank agreement of this size says that the closure statistic largely reads out how well each system’s bars are scaled. It is a statement about what the test responds to, and not evidence that the flagged systems are in error. The flagged systems do sit above the line, which is what any reproducible offset does: it inflates a closure residual without inflating the spread between repeats. Being reproducible across repeats that share a starting structure is what places a system above the line, and that is weaker than being in error.

Second, edges selected on one replicate are more inconsistent on the two held out than equal-size random selections are (mean gain in held-out reduced χ^2 , Stouffer $p = 1.3 \times 10^{-3}$, 1.3×10^{-3} , 4.4×10^{-3} over the three rotations), while a topology-only control built from curl-leverage is indistinguishable from random ($p = 0.21$ – 0.91). Each candidate statistic was calibrated against a no-error null before use and the realized level of every one is reported; two fired far above nominal and are not used. Only the consistency-gain statistic is specific to the flagged systems: an $|z|$ -rank-matched control drawn from unflagged systems reproduces the leverage-normalised residual signal but not the consistency gain. The effect is an aggregate one, significant individually in two of the five testable systems.

7 Computational details

Data. Every closure fit consumes the release’s combined per-replicate table directly, taking the per-leg complex and solvent ΔG and $d\Delta G$ columns and composing the edge value and its standard error from them; no network-level or cycle-closure correction is applied to an edge before it enters the fit. Figure 4 uses the `alchemtest` BACE1 relative-binding and benzene sets, from which forward and reverse work samples are available per λ window. The replicate validation, the cycle-closure test and every experiment on real binding data use the OpenFE IndustryBenchmarks2024 release [2], which ships three independent protocol repeats per transformation: 1145 edges across 49 benchmark systems. A system here is one of the benchmark’s named edge networks and not one protein, `jak2_set1` and `jak2_set2` being separate systems built on the same protein, so the system count exceeds the number of distinct proteins (Section 6.1). The 48 systems with at least one independent cycle, which between them hold 1143 of the 1145 edges, admit the closure test, and each counts as one hypothesis in the Benjamini–Hochberg correction; the single remaining system is acyclic and contributes the other two edges. The per-system calibration ratios of Figure 5B are pooled instead over the 34 systems that carry at least eight edges. Eight is the smallest edge count at which a ratio pooled against an $n = 3$ replicate spread is stable enough to plot on its own row; the other fifteen systems enter the aggregate ratio and the per-edge panel but are not shown individually. Grounding against measured affinity uses public ChEMBL IC₅₀ values. The downstream uses audited in Section 8 run instead on the public OpenFF protein–ligand benchmark [8], which aggregates the FEP+ and Merck data sets [24, 28] and carries curated per-edge experimental $\Delta\Delta G$ labels, under both-endpoint scaffold-disjoint out-of-distribution splits; interval sharpness, commit-to-synthesis selection, reward re-ranking and the commit-trust decision run on it, while edge ranking and the stopping decision run in controlled simulations. No free-energy simulation was run for this

work; all alchemical output is taken as published.

Per-edge variance. $\widehat{\Delta F}$ is obtained by solving Bennett’s estimating equation $S(\Delta F) = 0$ by scalar root-find, and its variance from the closed form B/I^2 of Theorem 3 evaluated at the root. The variance is therefore computed from the same simulation output as the estimate, at no label cost.

Cycle-closure test. Node potentials are fitted by generalized least squares on the signed incidence matrix of the perturbation network, whitened by the per-edge standard errors, so that the fit is a weighted least-squares problem with conductances $1/V_e^{\text{rep}}$. Writing N for that incidence matrix, to keep B for the sandwich meat of Theorem 3, the statistic is the reduced χ^2 of the residuals with $\text{dof} = E - \text{rank}(N)$. Per-system tail probabilities use the exact regularized upper incomplete gamma function rather than a Wilson–Hilferty approximation. Systems are flagged by Benjamini–Hochberg control of the false discovery rate at $\alpha = 0.05$ across systems. The population for the closure test is the 48 systems that carry at least three replicate-0 edges and at least one independent cycle, which between them hold 1143 of the benchmark’s 1145 edges.

The two baseline detectors. The fixed hysteresis cutoff flags a system when the largest absolute closure sum over a fundamental cycle basis exceeds 1.0 kcal/mol. That basis is the cotree of a spanning tree of the perturbation network, so it holds one cycle per degree of freedom of the closure fit, and the aggregation over them is a maximum. The rule consumes no per-edge uncertainty and applies no multiplicity correction. The fixed-se test is the closure fit above with one standard error per system, the median of that system’s per-edge reported values, substituted for every edge; its per-system tail probabilities pass through the same Benjamini–Hochberg control at $\alpha = 0.05$. The three detectors accordingly differ only in what they do with the per-edge bar: the cutoff ignores it, the fixed-se test replaces it by one number per system, and the calibrated test uses all of them.

Simulation protocols and model architectures. Two protocols underwrite numbers quoted above: the pure-null simulation that supplies every realized false-flag level, and the reproducible-systematic magnitude used as the label in detection comparisons. The amortized predictor used in the acquisition, stopping and reward experiments, whose predictive uncertainty $\sigma_{\text{total}} = \gamma \sqrt{\sigma_{\text{ens}}^2 + \sigma_{\text{ale}}^2}$ composes deep-ensemble epistemic spread with a benchmark aleatoric floor and is conformally scaled at miscoverage $\alpha = 0.1$, and the learned-variance baseline of Supplementary Figure S2 are separate and much smaller models. Both protocols, and the features, layer sizes, objectives and training budgets of both models, are specified in the Supporting Information.

Software. Python ≥ 3.11 with numpy 2.5.0, scipy 1.18.0, pymbar 4.0.3, torch 2.12.1, RDKit 2026.3.3 and matplotlib 3.11.0. The pipeline is deterministic: in the environment recorded here, re-running a figure’s command reproduces that figure’s rendered raster byte for byte, and this was checked on the figures that depend on ensemble training rather than assumed. Ensemble training runs full batch on CPU for a fixed number of epochs, each member is seeded from its own index, and the ligand ordering reaching the featurizer and the sampler is sorted rather than read out of a set, so no unrecorded source of variation enters the training path.

8 Limitations

The sandwich is aleatoric-only and conditional on correctly sampled ensembles. Systematic force-field bias lies outside it, and the out-of-distribution behaviour audited here rests on an ensemble plus a scalar aleatoric floor: there is no learned force-field-correction head, so the three-way decomposition of the synthetic illustration is not reproduced on real data.

The binding-edge se is validated against three independent replicates (Figure 5). In aggregate the replicate ratio is $\sim 1.41\times$ and the bar is conservative, but calibration is heterogeneous. Of the 34 systems carrying at least eight edges, 28 are conservative and six overconfident: **t4_lysozyme** $0.99\times$, **hif2a** and **bace** $0.96\times$, **faah** $0.90\times$, **brd4** $0.76\times$, and the protonation variant **bace_p3_arg368_in** $0.41\times$. That last is the only one of the six that is not marginal. Those replicates are seed and velocity repeats from identical starting coordinates, so the across-replicate SD is a lower bound on full reproducibility, excluding pose and preparation variance, and the measured conservatism may be closer to neutral against real reproducibility. The same restriction limits what the quality-control test detects: a pose or a protonation assignment carried unchanged into all three repeats reproduces exactly as physical error does, so the flags are reported as reproducible inconsistency throughout.

The test detects only *edge-level* inconsistency, and neither removal rule improves accuracy against experiment. Guided removal ties random on the four grounded systems (Stouffer $p = 0.48$; Figure S10). The influence rule does no better overall, doing measurable harm on **p38** and reaching a nominal improvement on **bace**, neither clearing the conjunctive criterion (Section 4; Figure 2C). Theorem 2 offers a structural reading of both nulls, bounded by the margin Section 4 measures and not established independently of it. The pre-registered falsifier for that reading is a guided removal beating its random null in a majority of grounded systems at a combined p below 0.05, which would show the visible part actionable; it is not met.

The measurement rests on four systems selected by the test, so it characterises those systems rather than the benchmark, and for **cdk8** it characterises less even than that. Its grounded sub-network is all 31 edges of **merck_cdk8**, the block whose own closure statistic is not significant on any replicate, so the evidence that flagged it lies outside the sub-network measured. **p38**’s grounded sub-network is the **jacs_p38** block, which is individually significant ($\chi^2_\nu = 2.63$, $p = 1.9 \times 10^{-4}$), and **hif2a** and **bace** are single-block systems.

The measurement also runs on a connectivity-based ChEMBL join that is stereo-blind, so enantiomer pairs sharing a 2D graph collapse to one experimental value. That inflates absolute MUE while cancelling in the guided-versus-random contrast, since both arms read the identical cached affinity. The experimental and annotation errors the join carries are per-ligand quantities entering the same unauditable subspace as force-field bias, so the measurement locates the error without attributing it.

The influence rule is a second pre-registered hypothesis on the data that returned the first null, so it is a second look and not independent confirmation. The theorem bounds what a network’s own edge values can support and says nothing against methods bringing information from outside the network. A prospective flag, repair and remeasure test with new molecular dynamics lies outside what public data can support.

The cross-replicate check shows that what is flagged reproduces across independent runs; it is not a false-positive-rate guarantee and does not certify the nominal Benjamini–Hochberg rate. Recall cannot be measured at all, there being no ground-truth error labels. Replacing each system’s reported standard error by its own replicate-measured calibration changes the flag set in both directions, by an amount depending on the granularity and the direction, from retaining two of the six to reaching sixteen. The arms, their realized false-flag probabilities and the single one that holds its level are tabulated in the Supporting Information, and none is reported here as a flag set.

The flag set is therefore sensitive to calibration error of the size actually present, in both directions, which limits how far any single flag list should be trusted.

Together with the median reduced χ^2 , which implies limited single-replicate power, that makes the flagged systems a floor rather than a prevalence estimate. **renin** carries the reproducible signature by the pooled-replicate ratio, its reduced χ^2 rising to 4.7 pooled from a per-replicate mean of 2.5. It clears the false-discovery threshold on two of the three replicates, though not on the first, which is the one the published list is drawn from. It is therefore a system that list misses rather than one the test misses, and **ciordia_retro** is a second such case.

A calibrated uncertainty was audited on six downstream uses: interval sharpness, edge ranking in active learning, commit-to-synthesis selection, reward re-ranking over a measured congeneric series, and two decision tasks, commit trust and calibrated stopping. The quantity under test there is not the per-edge sandwich bar of Section 5 but the predictive uncertainty of an amortized $\Delta\Delta G$ predictor, a conformally scaled combination of ensemble spread and an aleatoric floor; the Supporting Information defines it. It leads a fair baseline in none of the six, and on one it does harm: the calibrated reward lowers the true hit-rate of a generator on the single evaluable congeneric series, because the uncertainty there tracks chemical novelty rather than quality. On commit trust the physics σ matches post-hoc recalibration by temperature scaling [7] or isotonic regression [31] rather than beating it, and the separation in the stopping simulation is absent at the six per cent miscalibration a same-budget pooled estimator reaches. Calibration is a precondition for decisions of this kind rather than a multiplier on their quality. Supplementary Table S11 collects every effect against its baseline.

The stopping experiment is a controlled simulation, and the reward re-ranking audit rests on a single σ -varying target, so its conclusion that σ tracks novelty rather than quality is directional and should be tested across more chemotypes.

The cost of resolving the gradient part, one absolute measurement per unresolved dimension, is a worst case rather than an expectation. A bias smooth in chemical space is a far smaller object than one offset per ligand, and a noisy anchor resolves its dimension only to its own precision.

9 Conclusion

A perturbation network’s own edge values determine only the cycle part of its error. Fitting cycle closure against the precisions the free-energy estimator already reports turns a descriptive consistency check into a test with a stated null, orders networks by how far their own calculations disagree, and names in advance the edges that lie in no cycle and can carry no evidence. On these data the test catches nothing the fixed hysteresis cutoff misses: the two flag sets are nested and the head-to-head between them is a tie, so the advantage is structural, being per-edge adaptive thresholds, a null measured against seed and velocity repeats, and the observability map. The flags are not a repair list either. Removing the edges they name does not recover accuracy against experiment, and a rule built from the theorem does no better, because the component that sets accuracy against experiment lies in the gradient part, which no closure statistic reaches.

The test is therefore bounded twice: by the geometry of the network, and by a variance component its own null never sees, since that null is measured against repeats holding one pose and one system preparation fixed. Widening the reported errors by half already leaves nothing on this benchmark flagged, and half is less than published estimates of what preparation costs. Of the six systems flagged on the first replicate, one carries its flag from a naming collision created by the grouping the analysis chose, one is the weakest on every measure applied to it and drops when the replicates are pooled, and the membership of the set does not reproduce across the three repeats;

what survives all three is the ordering of systems and the per-edge residuals behind it. The practical use of cycle closure is accordingly as a calibrated alarm, with the error that moves a ranking to be sought in absolute measurements or independent preparations instead.

Supporting Information Available

Per-window overlap for the 19 real BACE1 adjacent- λ BAR windows; the sandwich standard error at the raw versus the effective sample count; the controlled overlap sweep of the closed form against learned-variance baselines; the Fisher-resistance correspondence between the sandwich precision and the conductance of a measurement graph; simulated active-learning acquisition policies and the knowledge-gradient score on gauge and decision-relevant contrasts; structural context for two flagged systems from public co-crystal structures; per-system curl-leverage and its relation to reproducible residuals; the change in unsigned error against experimental affinity after guided edge removal; detection AUC for three flagging rules; the per-system auditable dimension and detectable-shift resolution over the benchmark, the added ligand pairs that leave no bridge anywhere, and the influence-ranked repair race against random removal; the visible fraction recomputed on curated per-edge affinities over the whole benchmark, split by whether the closure test flags the system, with the node-keying convention settled by structure and the name-keyed value reported alongside; the label-noise floor beneath that fraction, with the exact annihilation of a per-ligand field by the projector and the per-system shrinkage each declared noise level would have to supply; the cross-release split of the multi-release systems and what it does and does not say about between-preparation variance; the architectures and training budgets of the amortized predictor and the learned-variance baseline, and the pure-null and reproducible-systematic simulation protocols; cycle-closure flagging under four per-edge σ assignments and across a swept uniform σ scale; per-system reduced χ^2 before and after inflating each system’s own bar; the true hit-rate of a trajectory-balance GFlowNet under three reward functions; a table collecting the effect of the decomposed σ across six downstream tasks; the replicate stability of the flag set, with per-system reduced χ^2 , degrees of freedom and Benjamini–Hochberg adjusted q -values tabulated for each of the three protocol repeats; the out-of-sample behaviour of the calibrated null, both as a prediction of each system’s closure statistic from replicates that do not enter it and as a held-out test of the edges it selects; conditional calibration and interval sharpness on real out-of-distribution data, the amortized reward experiment, and the calibrated stopping rule with its assumed-error sweep. (PDF)

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Author Contributions

N.L.P. is the sole author and is responsible for the conception and design of the study, the theory, the software, the analysis, the preparation of the figures, and the writing of the manuscript.

Conflicts of Interest

The author declares no competing financial interest.

Data and Software Availability

All data used in this work are public. The alchemtest BACE1 and benzene alchemical sets supply the work samples of Figure 4; the OpenFE IndustryBenchmarks2024 replicate set supplies the 1145 binding edges and their three protocol repeats; the OpenFF protein–ligand benchmark, which aggregates the FEP+ and Merck data sets, supplies the downstream audit tasks; and ChEMBL supplies the experimental affinities used for grounding. No free-energy simulation was run for this work; all alchemical output is taken as published. Two findings reported here were sent to the maintainers of that benchmark as public issues on its repository, <https://github.com/OpenFreeEnergy/IndustryBenchmarks2024>: the per-edge auditability and detectable-shift tables of Section 3 as issue 331 on 9 August 2026, and the `cdk8` ligand-name collision of Section 6 as issue 332 on 22 August 2026. On the second the maintainers confirmed the collision, reported that their own analyses were unaffected by it, and added a note to the release’s analysis documentation recording that ligand names are not unique across data sets. The first was closed as outside the scope of that repository. The BAR layer, the cycle-closure quality-control module, the theorem-verification and quality-control unit tests, and every figure-regeneration script are released as open source under a permissive licence, with each figure in this article regenerated by a single documented command. The code is available at <https://github.com/doctawho42/fep-surrogate-pipeline> under the MIT licence; the version from which every figure in this article was generated is tagged `v1.3.0`. The archive is deposited at <https://doi.org/10.5281/zenodo.22046692>, a concept identifier that resolves to the most recently deposited version.

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